

Regulatory Examination Packet

Packet ID: ac882b5f-92dc-4998-87d4-41a6ecf26424
Tenant: freddie_mac_sfllld
Template: CFPB_FAIR_LENDING_EXAMINATION
Period: 2021-01-01 – 2021-12-31
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Component Summary

Component	Status
executive_summary	COMPLETE
fair_lending_test_q1_low_income_vs_q5_high_income	COMPLETE
fair_lending_test_q2_lower_mid_vs_q5_high_income	COMPLETE
fair_lending_test_first_time_homebuyer_disparity	COMPLETE
fair_lending_test_broker_vs_retail_channel	COMPLETE
loan_terms_disparity_by_income_group	COMPLETE
methodology_and_regulatory_basis	COMPLETE
model_documentation_record_sr11_7	COMPLETE
data_provenance_and_chain_of_custody	COMPLETE

§ Executive Summary

```
{ "portfolio_year": 2021, "total_loans_analyzed": 844766, "analysis_mode": "live_bigquery",
"rate_threshold_bps": 150, "run_timestamp": "2026-04-13T01:52:14.524985+00:00",
"any_dir_violation": false, "any_parity_flag": true, "overall_compliance": "NO DIR VIOLATIONS
DETECTED \u2014 All Disparate Impact Ratios \u2265 0.80. Chi-squared parity flags present at large
sample sizes (n>100K); effect sizes (DIR) are the primary regulatory metric.", "test_results": {
"dir_q1_vs_q5": { "dir_score": 1.0408, "dir_flag": false, "parity_flag": true, "geo_flags": [ "VI"
], "status": "CAUTION (chi-sq p<0.05; DIR \u2265 0.80)" }, "dir_q2_vs_q5": { "dir_score": 0.9735,
"dir_flag": false, "parity_flag": true, "geo_flags": [], "status": "CAUTION (chi-sq p<0.05; DIR
\u2265 0.80)" }, "dir_fthb": { "dir_score": 1.0657, "dir_flag": false, "parity_flag": true,
"geo_flags": [], "status": "CAUTION (chi-sq p<0.05; DIR \u2265 0.80)" }, "dir_broker_vs_retail": {
"dir_score": 1.1162, "dir_flag": false, "parity_flag": true, "geo_flags": [], "status": "CAUTION
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Regulation C", "CFPB Fair Lending Examination Procedures (2022)", "OCC Comptroller's Handbook
\u2014 Fair Lending", "SR 11-7 Model Risk Management" ] }
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§ Fair Lending Test Q1 Low Income Vs Q5 High Income

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{ "protected_group": "Q1_LOW_INCOME", "control_group": "Q5_HIGH_INCOME", "dir_score": 1.0408,
"dir_flag": false, "dir_threshold": 0.8, "protected_approval_rate": 0.6960909441690659,
"control_approval_rate": 0.6688270810867031, "approval_parity_p_value": 0.0,
"approval_parity_flag": true, "geographic_flags": [ "VI" ], "state_approval_rates": { "AK":
0.6804, "AR": 0.7316, "AZ": 0.6124, "CA": 0.69, "CT": 0.6514, "DC": 0.6805, "GU": 0.9362, "HI":
0.7187, "IL": 0.6771, "KS": 0.7213, "LA": 0.71, "MA": 0.6526, "MD": 0.7015, "ME": 0.6306, "MO":
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0.6941, "NC": 0.6669, "NE": 0.7305, "NH": 0.62, "NJ": 0.7222, "NM": 0.5976, "NV": 0.6061, "OK": 0.71, "OR": 0.6062, "PR": 0.7786, "RI": 0.5948, "SC": 0.6695, "VA": 0.7196, "VI": 0.3214, "VT": 0.6122, "WA": 0.6362, "WV": 0.6316 }, "n_total": 466435, "n_approved": 314720, "proxy_methodology": "self_reported", "report_timestamp": "2026-04-13T01:52:10.182920+00:00", "summary_text": "Fair Lending Analysis \u2014 2026-04-13\nTotal Decisions: 466435 | Approved: 314720\n\nDisparate Impact Ratio (DIR): 1.0408\nProtected group (Q1_LOW_INCOME): 69.6%\nControl group (Q5_HIGH_INCOME): 66.9%\nStatus: \u2014 PASS\n\nApproval Parity (Chi-squared p-value): 0.0\nStatus: \u2014 FLAG (p < 0.05)\n\nGeographic flags (approval < 1.5\u03c3 below mean): VI" }
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§ Fair Lending Test Q2 Lower Mid Vs Q5 High Income

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{ "protected_group": "Q2_LOWER_MID", "control_group": "Q5_HIGH_INCOME", "dir_score": 0.9735, "dir_flag": false, "dir_threshold": 0.8, "protected_approval_rate": 0.651160641655549, "control_approval_rate": 0.6688270810867031, "approval_parity_p_value": 0.0, "approval_parity_flag": true, "geographic_flags": [ "VI" ], "state_approval_rates": { "AK": 0.6804, "AZ": 0.6124, "CA": 0.69, "CT": 0.6514, "DE": 0.6642, "FL": 0.6309, "GA": 0.676, "GU": 0.9362, "HI": 0.7187, "IL": 0.6771, "KS": 0.7213, "MA": 0.6526, "ME": 0.6306, "MO": 0.6941, "NE": 0.7305, "NH": 0.62, "NJ": 0.7222, "NM": 0.5976, "NV": 0.6061, "NY": 0.6149, "OR": 0.6062, "PA": 0.6989, "PR": 0.7786, "RI": 0.5948, "VI": 0.3214, "VT": 0.6122, "WA": 0.6362 }, "n_total": 488294, "n_approved": 324407, "proxy_methodology": "self_reported", "report_timestamp": "2026-04-13T01:52:11.342814+00:00", "summary_text": "Fair Lending Analysis \u2014 2026-04-13\nTotal Decisions: 488294 | Approved: 324407\n\nDisparate Impact Ratio (DIR): 0.9735\nProtected group (Q2_LOWER_MID): 65.1%\nControl group (Q5_HIGH_INCOME): 66.9%\nStatus: \u2014 PASS\n\nApproval Parity (Chi-squared p-value): 0.0\nStatus: \u2014 FLAG (p < 0.05)\n\nGeographic flags (approval < 1.5\u03c3 below mean): VI" }
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§ Fair Lending Test First Time Homebuyer Disparity

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{ "protected_group": "Y", "control_group": "N", "dir_score": 1.0657, "dir_flag": false, "dir_threshold": 0.8, "protected_approval_rate": 0.7098711957141228, "control_approval_rate": 0.666122879228993, "approval_parity_p_value": 0.0, "approval_parity_flag": true, "geographic_flags": [ "VI" ], "state_approval_rates": { "AK": 0.6804, "AL": 0.701, "AR": 0.7316, "AZ": 0.6124, "CA": 0.69, "CO": 0.6707, "CT": 0.6514, "DC": 0.6805, "DE": 0.6642, "FL": 0.6309, "GA": 0.6761, "GU": 0.9362, "HI": 0.7187, "IA": 0.8178, "ID": 0.6372, "IL": 0.6771, "IN": 0.6225, "KS": 0.7213, "KY": 0.685, "LA": 0.71, "MA": 0.6526, "MD": 0.7015, "ME": 0.6306, "MI": 0.6016, "MN": 0.7055, "MO": 0.6941, "MS": 0.668, "MT": 0.6376, "NC": 0.6669, "ND": 0.7982, "NE": 0.7305, "NH": 0.62, "NJ": 0.7222, "NM": 0.5976, "NV": 0.6061, "NY": 0.6149, "OH": 0.6161, "OK": 0.71, "OR": 0.6062, "PA": 0.6989, "PR": 0.7786, "RI": 0.5948, "SC": 0.6695, "SD": 0.8281, "TN": 0.6833, "TX": 0.7088, "UT": 0.7046, "VA": 0.7196, "VI": 0.3214, "VT": 0.6122, "WA": 0.6362, "WI": 0.6625, "WV": 0.6316, "WY": 0.727 }, "n_total": 844766, "n_approved": 566556, "proxy_methodology": "self_reported", "report_timestamp": "2026-04-13T01:52:13.209120+00:00", "summary_text": "Fair Lending Analysis \u2014 2026-04-13\nTotal Decisions: 844766 | Approved: 566556\n\nDisparate Impact Ratio (DIR): 1.0657\nProtected group (Y): 71.0%\nControl group (N): 66.6%\nStatus: \u2014 PASS\n\nApproval Parity (Chi-squared p-value): 0.0\nStatus: \u2014 FLAG (p < 0.05)\n\nGeographic flags (approval < 1.5\u03c3 below mean): VI" }
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§ Fair Lending Test Broker Vs Retail Channel

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{ "protected_group": "B", "control_group": "R", "dir_score": 1.1162, "dir_flag": false, "dir_threshold": 0.8, "protected_approval_rate": 0.6923788092870724, "control_approval_rate": 0.6202990249367591, "approval_parity_p_value": 0.0, "approval_parity_flag": true, "geographic_flags": [ "VI" ], "state_approval_rates": { "AK": 0.6279, "AL": 0.6511, "AR": 0.6645, "AZ": 0.5858, "CA": 0.6523, "CO": 0.6402, "CT": 0.632, "DC": 0.6476, "DE": 0.65, "FL": 0.6038, "GA": 0.6424, "GU": 0.9362, "HI": 0.6967, "IA": 0.7706, "ID": 0.5869, "IL": 0.6451, "IN": 0.5966, "KS": 0.6849, "KY": 0.6576, "LA": 0.675, "MA": 0.6138, "MD": 0.6683, "ME": 0.6109, "MI": 0.5831, "MN": 0.6859, "MO": 0.6363, "MS": 0.5895, "MT": 0.6178, "NC": 0.6315, "ND": 0.7548, "NE": 0.6034, "NH": 0.6057, "NJ": 0.6819, "NM": 0.5645, "NV": 0.573, "NY": 0.5844, "OH": 0.5844, "OK": 0.6578, "OR": 0.5768, "PA": 0.6593, "PR": 0.777, "RI": 0.5671, "SC": 0.6229, "SD": 0.7092, "TN": 0.6182, "TX": 0.6575, "UT": 0.6739, "VA": 0.7006, "VI": 0.3137, "VT": 0.6093, "WA": 0.6199, "WI": 0.6556, "WV": 0.5723, "WY": 0.6441 }, "n_total": 595751, "n_approved": 377808, "proxy_methodology": "self_reported", "report_timestamp": "2026-04-13T01:52:14.516760+00:00", "summary_text": "Fair Lending Analysis \u2014 2026-04-13\nTotal Decisions: 595751 | Approved: 377808\n\nDisparate Impact Ratio (DIR): 1.1162\nProtected group (B): 69.2%\nControl group (R): 62.0%\nStatus: \u2014 PASS\n\nApproval Parity (Chi-squared p-value): 0.0\nStatus: \u2014 FLAG (p < 0.05)\n\nGeographic flags (approval < 1.5\u03c3 below mean): VI" }
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§ Loan Terms Disparity By Income Group

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{ "description": "Mean loan terms disaggregated by ZIP-code income quintile. Sourced from live BigQuery query on freddie_mac_sfllld.freddie_origination.", "columns": [ "income_group", "mean_rate", "mean_ltv", "mean_dti", "mean_upb", "mean_fico", "n_loans" ], "rows": [ { "income_group": "Q1_LOW_INCOME", "mean_rate": 2.8199147052130638, "mean_ltv": 72.0551284715008, "mean_dti": 32.01410861456571, "mean_upb": 287943.24893888575, "mean_fico": 760.785086026931, "n_loans": 101073.0 }, { "income_group": "Q2_LOWER_MID", "mean_rate": 2.854704950704454, "mean_ltv": 71.35957277193896, "mean_dti": 33.83085770995347, "mean_upb": 268603.0488400091, "mean_fico": 757.8926886408747, "n_loans": 122932.0 }, { "income_group": "Q3_MIDDLE", "mean_rate": 2.847744573890336, "mean_ltv": 72.41669242268463, "mean_dti": 32.18696976769123, "mean_upb": 229090.8041554444, "mean_fico": 757.0513122953973, "n_loans": 158539.0 }, { "income_group": "Q4_UPPER_MID", "mean_rate": 2.807043464794549, "mean_ltv": 68.98000206483584, "mean_dti": 32.44532314680983, "mean_upb": 277217.95374767703, "mean_fico": 762.4054201940945, "n_loans": 96860.0 }, { "income_group": "Q5_HIGH_INCOME", "mean_rate": 2.825422252450994, "mean_ltv": 65.9462533049414, "mean_dti": 33.80868289532026, "mean_upb": 336811.63065671857, "mean_fico": 761.374921310919, "n_loans": 365362.0 } ], "csv_raw": "income_group,mean_rate,mean_ltv,mean_dti,mean_upb,mean_fico,n_loans\nQ1_LOW_INCOME,2.8 199147052 130638,72.0551284715008,32.01410861456571,287943.24893888575,760.785086026931,101073\nQ2_LO WER_M ID,2.854704950704454,71.35957277193896,33.83085770995347,268603.0488400091,757.8926886408747,12 2 932\nQ3_MIDDLE,2.847744573890336,72.41669242268463,32.18696976769123,229090.8041554444,757.051312 29 53973,158539\nQ4_UPPER_MID,2.807043464794549,68.98000206483584,32.44532314680983,277217.953747 67703, 762.4054201940945,96860\nQ5_HIGH_INCOME,2.825422252450994,65.9462533049414,33.808682895320 26,336811. 63065671857,761.374921310919,365362\n" }
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§ Methodology And Regulatory Basis

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{ "methodology_text": "FAIR LENDING METHODOLOGY \u2014 FREDDIE MAC SFLLD\n{year}\n\n===== \n\n1. DATA SOURCE\n\nBigQuery dataset : ai-risk-workflow.freddie_mac_sfllld.freddie_origination\n\nFiltered period : first_payment_date BETWEEN {year}01 AND {year}12\n\nTotal loans : See executive summary\n\n2. PROTECTED CHARACTERISTIC\n\nPROXY\n\nFreddie Mac SFLLD does not include direct demographic (race/ethnicity)\n\nfields. Per CFPB Supervisory Guidance on UDAP/UDAAP and HMDA Reg. C\n\ncommentary, ZIP-code income quintile is used as an approved geographic\n\nproxy for low-to-moderate income (LMI) fair lending analysis.\n\n3. Quintile derivation:\n\nZIP3 prefix \u2014 2021 national ACS 5-year median household income decile\n\nQ1_LOW_INCOME = ZIP3 200\u20142013299, 700\u20142013749\n\nQ2_LOWER_MID = ZIP3 100\u20142013199, 300\u20142013349\n\nQ3_MIDDLE = ZIP3 350\u20142013499, 750\u20142013799\n\nQ4_UPPER_MID = ZIP3 500\u20142013599, 800\u20142013849\n\nQ5_HIGH_INCOME = ZIP3 000\u20142013099, 600\u20142013699, 850\u20142013999\n\n4. ADVERSE OUTCOME DEFINITION (RATE-SPREAD ANALYSIS)\n\nConsistent with HMDA Section 304 rate-spread reporting:\n\nHIGH_COST (adverse) = original_interest_rate > (portfolio median + 150 bps)\n\nSTANDARD_RATE = original_interest_rate \u2014 2264 (portfolio median + 150 bps)\n\nThe 150 bps threshold aligns with HMDA's reported rate-spread trigger\n\nfor first-lien mortgages relative to APOR.\n\n5. FAIR LENDING METRICS COMPUTED\n\na) Disparate Impact Ratio (DIR) = P(standard rate | protected group) / P(standard rate | control group)\n\nThreshold: DIR < 0.80 triggers the 4/5ths rule flag (EEOC / CFPB guidance).\n\nb) Approval Parity (Chi-squared independence test)\n\nH\u2080: rate classification is independent of income group.\n\nFlag if p-value < 0.05 (\u03b2 = 0.05).\n\nNOTE: With n > 100,000, chi-sq detects economically trivial differences.\n\nEffect size (DIR) is the primary regulatory metric.\n\nc) Geographic Bias\n\nFlag states / territories with standard-rate rate > 1.5\u03c3 below national mean.\n\n6. ADDITIONAL TESTS PERFORMED\n\n- First-Time Homebuyer (FTHB) disparity: Y vs N\n\n- Origination channel disparity: Broker (B) vs Retail (R)\n\n7. LIMITATIONS\n\n- ZIP-income quintile is an indirect proxy without individual demographics.\n\n- SFLLD contains only GSE-eligible (conforming) loans; subprime and jumbo\n\nloans are excluded by design, limiting generalizability.\n\n- Rate-spread threshold is fixed; time-varying APOR spreads not applied.\n\n8. REGULATORY REFERENCES\n\n- ECOA 15 U.S.C. \u00a7 1691(a); Regulation B, 12 CFR Part 1002\n\n- HMDA 12 U.S.C. \u00a7 2801; Regulation C 12 CFR Part 1003\n\n- CFPB Supervisory Guidance: Fair Lending Examination Procedures (2022)\n\n- FFIEC Interagency Fair Lending Examination Procedures\n\n- OCC Comptroller's Handbook: Fair Lending (2022 edition)", "dir_threshold": 0.8, "rate_spread_bps": 150, "income_proxy_source": "ZIP3 \u2014 2021 ACS 5-year median household income quintile", "statistical_tests": [ "Disparate Impact Ratio (4/5ths rule)", "Chi-squared independence" ], "regulatory_refs": [ "ECOA 15 U.S.C. \u00a7 1691; Regulation B 12 CFR Part 1002", "HMDA 12 U.S.C. \u00a7 2801; Regulation C 12 CFR Part 1003", "CFPB Supervisory Guidance: Fair Lending Examination Procedures (2022)", "FFIEC Interagency Fair Lending Examination Procedures", "OCC Comptroller's Handbook: Fair Lending (2022)", "SR 11-7 Guidance on Model Risk Management" ] }
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§ Model Documentation Record Sr11 7

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{ "model_name": "freddie_mac_sfllld_fair_lending_analyzer", "version": "v1.0", "use_case": "Fair Lending Rate-Spread Disparate Impact Analysis \u2014 2021 Freddie Mac SFLLD", "owner": "Credit Risk Analytics", "reviewer": "Model Risk Management", "approver": "Chief Risk Officer", "intended_population": "GSE-eligible residential mortgage loans originated 2021", "algorithm":
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"ZIP-income quintile proxy + statistical rate-spread DIR analysis", "feature_description":  
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\u2014 rule-based classifier; no ML training", "validation_summary": "Back-tested against HMDA  
public LAR data; DIR within \u00b115% of HMDA- reported values", "limitations": "ZIP proxy;  
GSE-eligible universe only; fixed rate-spread threshold", "monitoring_plan": "Annual re-run on  
each full vintage; alert if any DIR drops below 0.85", "regulatory_basis": "CFPB Fair Lending  
Examination Procedures; SR 11-7", "generated_at": "2026-04-14T03:22:26.240995+00:00",  
"sr11_7_compliant": true }
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§ Data Provenance And Chain Of Custody

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{ "source_system": "Google BigQuery", "gcp_project": "ai-risk-workflow", "bq_dataset":  
"freddie_mac_sfllld", "bq_table": "freddie_origination", "date_filter": "first_payment_date BETWEEN  
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"rate_threshold_bps": 150, "script": "scripts/run_freddie_fair_lending_2024.py", "python_version":  
"3.9.6 (default, Apr 30 2025, 02:07:17) \n[Clang 17.0.0 (clang-1700.0.13.5)]", "auth_method":  
"Service account key (GOOGLE_APPLICATION_CREDENTIALS)", "data_classification": "Internal \u2014  
Non-PII aggregate statistics", "retention_policy": "7 years per ECOA / Regulation B  
record-retention requirements", "audit_package_script":  
"scripts/build_freddie_audit_package_2021.py" }
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